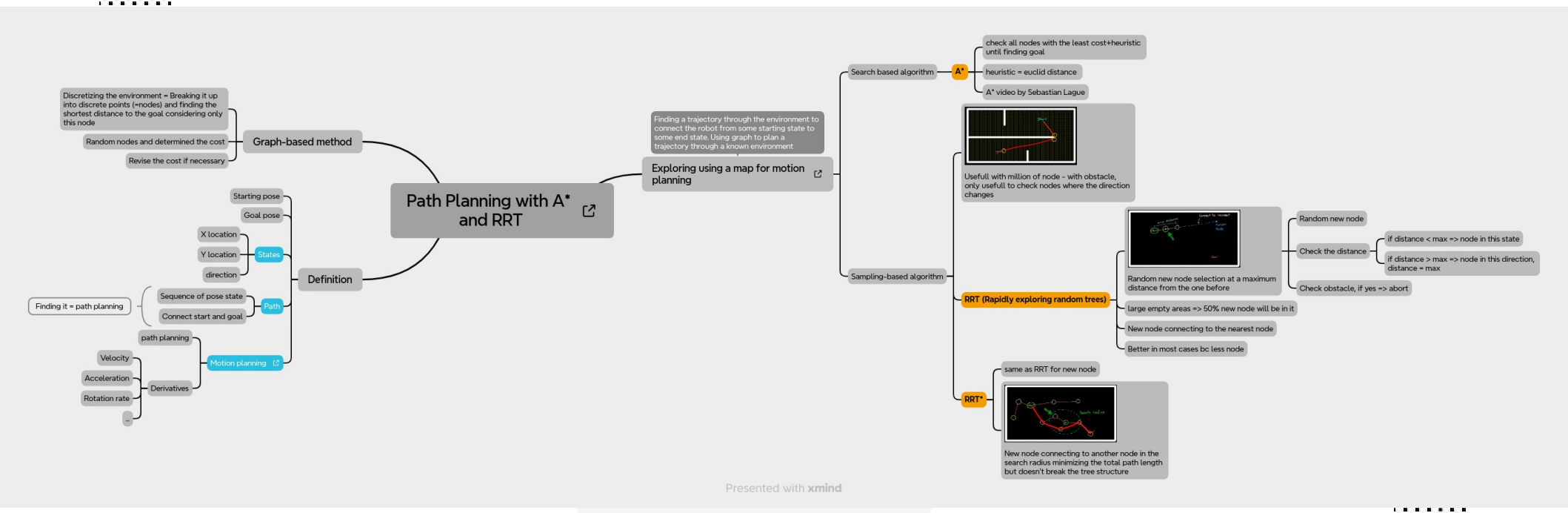
Path planning algorithm

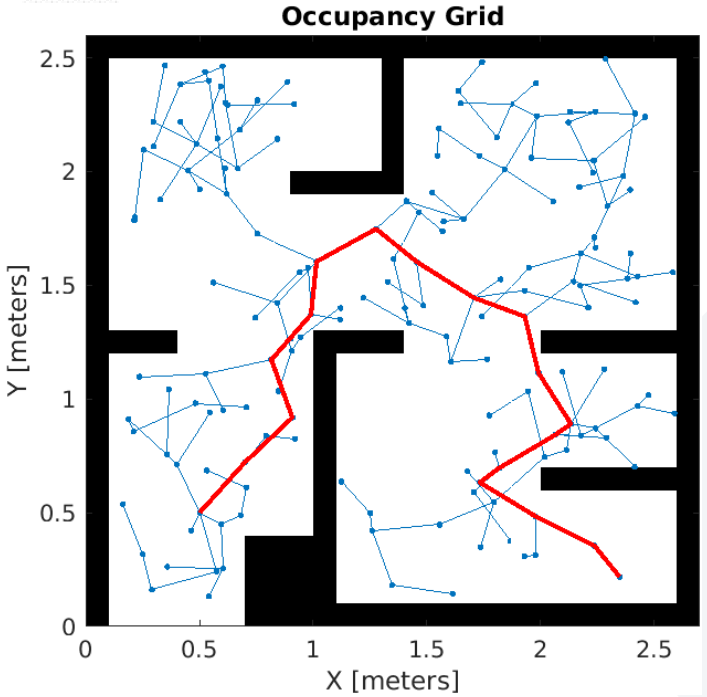
Plan

04 July 2025

Path planning with A* and RRT



Path planning with A* and RRT

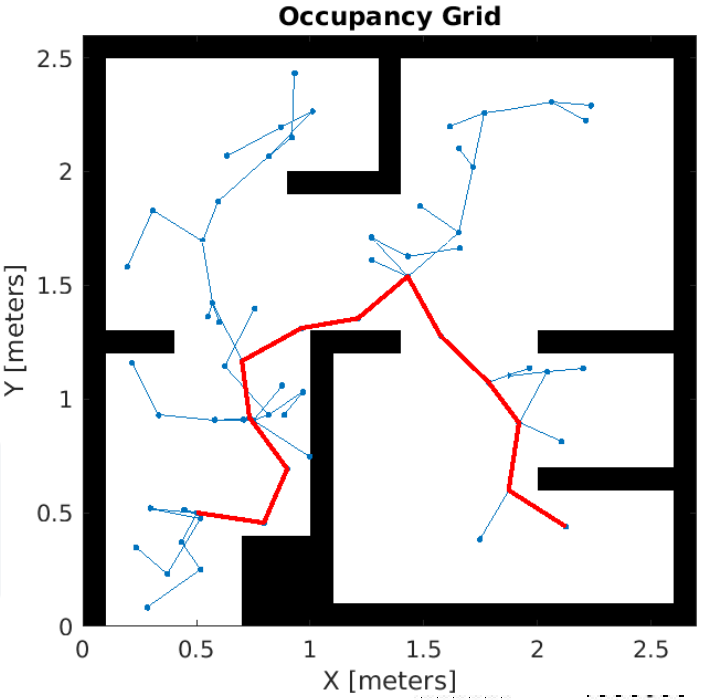


RRT implementation

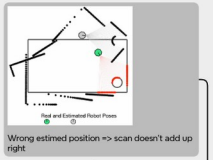
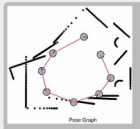
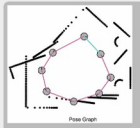
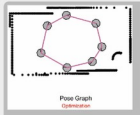
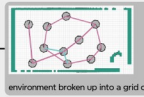
Improve to RRT*

Change parameters

Execution is too long

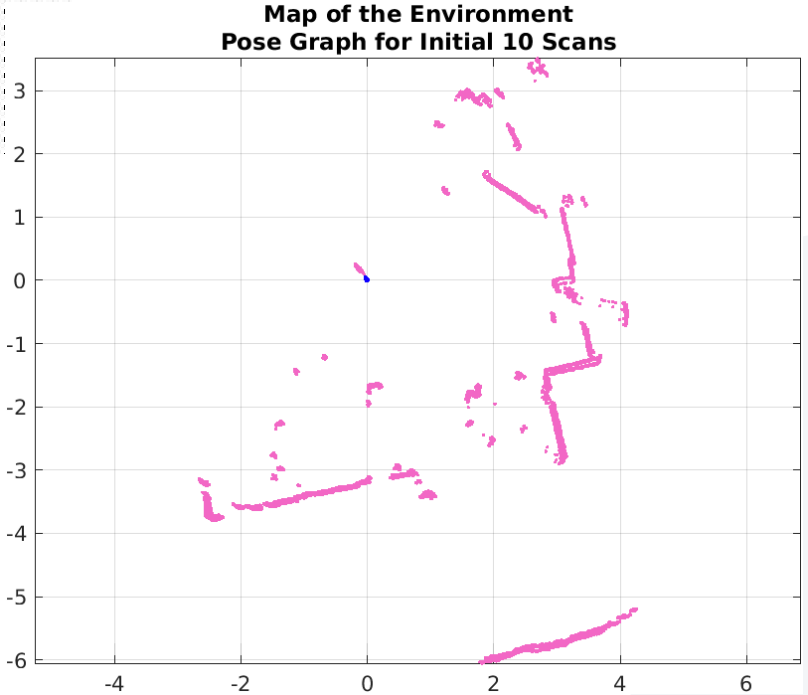


SLAM Algorithm

- SLAM Algorithm
  - Build a map while estimate a vehicle's pose
  - Understanding SLAM Using Pose Graph Optimization (Simultaneous Localization And Mapping)
    - Definition
      - Map
        - Model of the environment
      - Vehicle's pose
        - location
        - orientation
    - Autonomous vehicle (Robot)
      - Lidar
        - Dead reckon its relative position over time
          - Odometry (Wheel encoders)
    - Implémentattion
  - Localisation and estimed pose
    - Filtering
      - EKF
      - Particle
      - State is estimates "on the go" with latest measurements
    - Smoothing
      - Pose graph optimization
      - Full trajectories are estimated using complete set of measurements
    - No uncertainty or errors
      - Scan
      - Move
      - 
    - Lidar accurate, large uncertainty in odometry
      - 
        - Wrong estimed position => scan doesn't add up right
      - Build a Pose graph
        - On a graph put the estimated pose and scan at each step
        - 
          - Add a constraints linking the 2 poses, lake a rubber bar. The second pose can be moved but the rubber bar always put it back to the way it was. The strength of the rubber bar depends on how confident we are on the distance estimate
        - Estimed pose = node, constraints (rubber bar) = edge
        - 
          - The last estimate position is suppose to be the same as the first one, so we close the loop by adding a new constraint, a bar with a nominal length of 0. The strength of this rubber bar depends on how confident you are in your external measurements and your ability to associate the two sets of data as the same feature.
        - 
          - The loop closure add tension into the graph. The blue bar want to put this two nodes together while the purple bars all want to keep the relative distances the way they are.
        - To improve the system, add new nodes
        - It is better to miss a real loop closure than to add a false one
  - Mapping
    - Binary Occupancy Grid
      - 
        - environment broken up into a grid of cells
      - Occupied => 1 (green)
      - Empty => 0 (white)

Presented with xmind

SLAM Algorithm



- Retrieval of a dataset from MIT Stata Center project
- Parsing the dataset
- Combination of the first 10 scans

- Complete execution too long

Future goals

Matlab simulation

Algorithms implementation RRT, RRT*

SLAM implementation

Improve path-planning algorithms

Real-time robot navigation

Scan and map the environment (SLAM) using a lidar detector

Move from point A to B while avoiding obstacles



Transition to C++

Implement complex algorithms with C++

Execution faster, use of microcontrollers

Moving obstacle

Find documentation and implement algorithm for real time scan to avoiding new obstacle